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UNIVERSITÀ DI ROMA

# Employing Explainability of GNNs for Correlating COPD with Genetics and Lifestyle


Facoltà di Ingegneria dell'Informazione, Informatica e Statistica
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*To the wonderful educators that led me up here,*
*to the spirit of curiosity,*
*and to my dear friends and family.*

# Abstract

In this work, the correlation between chronic obstructive pulmonary disease (COPD), lifestyle (in particular, smoking) and genetics was studied by analysing how an ensemble of Graph Convolutional Networks (GCNs) classified gene co-expression networks, employing the GNNExplainer algorithm.
Four models were trained, each performing binary classification between two of the four possible cohorts - positive smokers, positive non-smokers, negative smokers, negative non-smokers. The models relied only on the correlation coefficients of co-expression networks, with no further biological information employed in the training or explanation phase. The chosen architectures showed high performances on the task (99% of accuracy on test set) and the classification explanations for each cohort were coherent among the different models.
The GCNs tended to base predictions on gene sets that have already been correlated with COPD in the literature, at relevant significance level ($\alpha = 0.05$) and both in the case of positivity and negativity to the disease.

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# Chapter 1

# Introduction

CHRONIC obstructive pulmonary disease (COPD) is an airway disease characterized by lung airflow limitation and tissue destruction. It can result in chronic bronchitis, with airway inflammation and scarring, and it is a potential driver of tumour maturation, as $0.8 - 1.7\%$ of COPD patients develop cancer every year.[1]
The progression of the disease causes imbalances in proteinase/anti-proteinase activity and oxidant/anti-oxidant balance, leading to the formation of emphysema. Emphysema is the abnormal enlargement of the air spaces located peripherally from the terminal bronchioles, with destruction of the walls of these structures [1].
COPD is the fourth cause of death around the world [2].

In developed countries, tobacco smoking accounts for 70% of the cases [2], and the inhalation of irritants, such as air pollutants and chemicals, is believed to be associated with it [1].
Numerous genetic factors are associated to COPD, the most well-known being gene polymorphisms in molecules like tumour necrosis factor, or antioxidant gene dependence, which in turns correlates race and age with disease severity and phenotype.
Epigenetics is the study of how behavioural and environmental factors contribute to (reversible) changes in an individual's genetics. Methylation plays a critical role in COPD development, and its epigenetic mechanism can be altered by cigarette smoking [1].

Genetic information has a primal importance in pathology. The genetics of the immune system influences the response to a disease; genetic disorders account for 80% of rare disorders, and all disease processes are generally influenced by a multitude of genetic characteristics [3].
Gene expression is the process by which the information encoded in a gene is turned into a function [4]. Gene expression profiling is the measurement of the activity, or expression, of a high number of genes at once.
Expression data can be represented as gene co-expression networks, which are graphs where edges represent significant co-expression relations.
In pathology, the analysis of gene co-expression networks is essential to recognize the interplay of multiple molecular processes in expressing the pathophenotype: characterizing the network as a whole is essential for understanding disease complexity [5].

RNA and DNA have been modelled for a long time as graphs/networks, and gene co-expression networks are an example of this. A number of network science [6], machine learning and deep learning techniques have been applied in biosciences [7]. This has been greatly enhanced by advancements in explainable AI methods [8].

In this work, a study of the autonomous extraction of biologically relevant features from a deep learning model was carried out. In particular, a graph neural network (GNN) model was trained to perform a graph classification task on COPD gene co-expression networks. The task consisted in classifying whether the gene co-expression network was formed from positive or negative patients, and whether such patients were smokers or not.
After this, a gradient-based explainer algorithm (GNNExplainer) performed an optimization task on the model predictions in order to capture which genes were the most influential for each classification. The distributions of these gene sets were then analysed in order to extract the most consistently influent genes, and then confronted with the existing COPD genetic literature.

## 1.1 Outline

In chapter 2, similar approaches to the task are considered in order to motivate the considered one. Following it, in chapter 3 the raw dataset is presented, as well as the pre-processing phase. The deep learning methodology (model-explainer pipeline) applied is described in chapter 4, while a discussion of its results, both from a purely statistical and biological standpoint, is carried out in chapter 5. Conclusions and possibilities for future works are outlined in chapter 6.

# Chapter 2

# Related works

The work presented was motivated by a number of approaches that employed the possibility of

1. Using a deep learning model to perform a classification task;
2. Using an explainer algorithm to interpret the predictions of the deep learning model.

Some relevant works are presented here, and then the proposed method is described in comparison with them.

## 2.1 Node-informed models

In *XGDAG: explainable gene–disease associations via graph neural networks* [9], the general outline presented above was followed. The task was gene-disease association in the context of protein-protein interaction (PPI) networks; thus, it was a node classification task on a graph, each graph being the PPI network of a given disease.
Node labeling relies heavily on the domain context. Indeed, not only the genes that were known to be associated with the given disease were labeled as positive (P), but, through NIAPU - a Markovian diffusion process algorithm for gene discovery [10] - the other nodes, given the positively-labeled genes and the graph topology, were additionally labeled as likely positive, weakly negative, likely negative or reliably negative.

The model learnt to predict these five different node classes. In the context of explanations, each positive node classification was supported by a subgraph of adjacent nodes, which may have had different labels. Among these, the ones considered relevant for gene discovery were only those with the likely positive label; moreover, they were ranked considering how many explanation subgraphs they belonged to and the score given by the explainer algorithm.

Thus, the approach consisted in heavily biasing the gene classification of the model and its explanation towards known gene discovery mechanisms.

## 2.2 COPD-related works

The only work applying GNNs and explainer algorithms on COPD transcriptomics (the study of RNA transcripts) is [11]. It employed a multi-omic approach;

the graph was built as a protein-protein interaction network, encoding already known interaction information; on top of that, gene and protein expression data were used as signals of graph nodes in the PPI network. This was done by convoluting these signals with the graph convolutional layers.
In particular, the two types of information (from genes and from proteins) were used in two different, separate graphs, that underwent several graph convolution-pooling layers before being reunited with a fully connected layer.
The final task was graph classification of the protein-protein interaction networks, and the relevant genes were extracted from model classifications using SHAP [12].

In this work, the model's classifications were performed on a graph topology built on known protein-protein interactions. It was then considered to be relevant, for the work that will be presented in the next chapters, to assess the performance and behaviour of a less-informed model.

## 2.3 Work proposal

The aim of this work was developing a domain-agnostic model, whose classifications would then be analysed by a model-agnostic explainer algorithm. In this way, the autonomous potential of GNN to model such a complex disease was observed.
The model is domain-agnostic in the sense that the only information received is the one coming from gene expression correlation, and the graph construction was not biased in any way towards the reflection of any biologically known mechanism, nor was the node features construction, as it was random.
The chosen explainer is model-agnostic since it employed the model entirely as a black box, without relying on informed graph labeling (as it was proposed in [9].)
Further details for both aspects can be found in the respective sections in chapter 4.

# Chapter 3

# Dataset analysis

## 3.1 Overview

The considered dataset was collected by the Lung Tissue Research Consortium (LTRC). It contains gene expression data for non-emphysema predominant COPD patients and controls, separating smoker and non-smoker subjects. Gene expression profiling measures mRNA (messenger RNA) levels; the collection process employed RNA-seq [13].

The dataset consists of four cohorts:

| | |
|---|---|
| $\mathcal{C}_1$: smoker, positive<br>$\lvert\mathcal{C}_1\rvert = 206$ | $\mathcal{C}_2$: smoker, negative/control<br>$\lvert\mathcal{C}_2\rvert = 929$ |
| $\mathcal{C}_3$: non-smoker, positive<br>$\lvert\mathcal{C}_3\rvert = 236$ | $\mathcal{C}_4$: non-smoker, negative/control<br>$\lvert\mathcal{C}_4\rvert = 574$ |

Each of the 4 cohorts $\mathcal{C}_i$ has $|\mathcal{C}_i| \times 3035$ entries, with 3035 being the number of genes considered (which is fixed for all cohorts).

## 3.2 Preprocessing

For each of the 4 cohorts, a sampling procedure was adopted in order to **create a co-expression network dataset.**
For each cohort $i$, $n_i$ patients (with $n_i < |\mathcal{C}_i|$) were sampled at random $m$ times, thus creating $m$ expression sub-datasets of length $n_i$, for each of the 4 cohorts. Then, correlation networks were created from these sub-datasets: a simple Pearson correlation coefficient was used, with the correlation coefficient being defined as

$$p_{xy} = \frac{\sum_{j=1}^{n_i}(x_j - \overline{x})(y_j - \overline{y})}{\sqrt{\sum_{j=1}^{n_i}(x_j - \overline{x})^2 \sum_{j=1}^{n_i}(y_j - \overline{y})^2}} \tag{3.1}$$

with $x, y$ being genes, $x_j, y_j$ denoting the expression of them in patient $j$, and $\overline{x}$ and $\overline{y}$ being the respective expression averages.
Every edge of the co-expression network whose correlation was below 0.5 in absolute value was dropped. This allowed a sparser representation of the graph (this was mainly due to computational and memory reasons). Formally, each co-expression network is defined as a graph $G_l = (V, E_l)$, with

$$V = \{\text{The set of all genes}\}, \qquad \text{with } |V| = 3035$$

$$E_l = \{\{x, y\} \mid |p_{xy}^{(l)}| \geq 0.5\}$$

with $p_{xy}^{(l)}$ being the Pearson correlation coefficient calculated on the associated $l$-th subdataset (note how edges were modelled as sets, since correlation is, inherently, undirected). Thus, exactly $m$ co-expression networks were formed for each cohort.

Each of the four cohort datasets were combined in pairs to create 4 new datasets, where each data point is a correlation network:

1. **COPD**: a dataset with $m$ co-expression networks created from COPD, smoker gene profiles (class 1), and $m$ co-expression networks created from COPD, non-smoker gene profiles (class 0), for a total of $2m$ co-expression networks;

2. **Control**: a dataset with $m$ co-expression networks created from control, smoker gene profiles (class 1), and $m$ co-expression networks created from control, non-smoker gene profiles (class 0), for a total of $2m$ co-expression networks;

3. **Smokers**: a dataset with $m$ co-expression networks created from COPD, smoker gene profiles (class 1), and $m$ co-expression networks created from control, smoker gene profiles (class 0), for a total of $2m$ co-expression networks;

4. **Non-smokers**: a dataset with $m$ co-expression networks created from COPD, non-smoker gene profiles (class 1), and $m$ co-expression networks created from control, non-smoker gene profiles (class 0), for a total of $2m$ co-expression networks.

These are the names which the datasets, and the models trained on such datasets, will be referred with, and they will be denoted by a **bold** name (e.g. **smokers** to mean COPD smoker vs control smoker gene co-expression network dataset) in text, and $\mathcal{D}$ in formulae, with $\mathcal{D}_{\mathcal{C}}$ indicating a specific dataset class (for example, class 0 of **smokers**).

Since model ensembling was employed, in order to make model predictions and explanations independent from the specific sampling, the whole process was repeated $N_{models}$ times.
An overview of the whole data pre-processing procedure can be seen in algorithm 1. Note that this procedure needed to be repeated for training, validation and test independently, on separate portions of the dataset; such division was not specified in the algorithm to declutter the notation.

| Value/Set | Training | Validation | Test |
|---|---|---|---|
| $n_i$ (sampled data points) | $\lvert\mathcal{C}_i\rvert/10$ | | |
| $m$ | 512 | 64 | 64 |

**Table 3.1.** Dataset dimensions.

**Algorithm 1** Data pre-processing

**Input:** $N_{models},\ M,\ \mathcal{C}_i,\ n_i; \quad 1 \leq i \leq 4$
  **for** $n = 1 \ldots N_{models}$ **do**
    **for** $m = 1 \ldots M$ **do**
      **for** $i = 1 \ldots 4$ **do**
        From $\mathcal{C}_i$, sample at random $n_i$ patients and assign it to $patients[i]$
        Form $network[i]$ by correlating $patients[i]$
      **end for**
      $copd[n] \leftarrow copd[n] \cup \{network[1]\} \cup \{network[3]\}$
      $control[n] \leftarrow control[n] \cup \{network[2]\} \cup \{network[4]\}$
      $smokers[n] \leftarrow smokers[n] \cup \{network[1]\} \cup \{network[2]\}$
      $non - smokers[n] \leftarrow non - smokers[n] \cup \{network[3]\} \cup \{network[4]\}$
    **end for**
    Randomly permute $copd[n]$
    Randomly permute $control[n]$
    Randomly permute $smokers[n]$
    Randomly permute $non - smokers[n]$
  **end for**

# Chapter 4

# Methodology

## 4.1 Graph Convolutional Networks

The task dealt with classification, in an inductive setting, on graphs. It was natural to apply graph neural networks to model the task, which, in particular, was graph classification: given one of the four aforementioned datasets, and a graph from such dataset, the model needed to predict if it belonged to the first or the second class.
For example, if the **non-smoker** dataset was picked, it would classify whether the considered co-expression network was calculated from COPD patients or from control gene profiles.

The architecture of the final model was a **graph convolutional network** (GCN). Its name is due to the affinity with Convolutional Neural Networks, in the sense that just as a convolutional filter is swept over each pixel of an image, which can be seen as an aggregation operation from neighbouring nodes, GCNs apply an aggregation operation to the neighbourhood of each node, too. This is also called message-passing neural network [14].
The message-passing scheme is summarized in algorithm 2. Note that the neighbours of a node $v \in V$ are defined as

$$\mathcal{N}(v) = \{w \mid w \in V \ \wedge \ \{w, v\} \in E\} \tag{4.1}$$

**Algorithm 2** Generic message-passing neural network (without non-linearities)

**Input:**
$G = (V, E)$ (undirected),
$L$ the number of layers,
Node embeddings $\{h_v^{(0)} = x_v \in \mathbb{R}^{n_0}\}$,
Aggregate$^{(l)}(\cdot)$ functions, with $l = 0 \dots L-1$
**for** $l = 0 \dots L-1$ **do**
  $h_v^{(l+1)} \leftarrow \text{Aggregate}^{(l)}(\{h_w^{(l)} : \ w \in \mathcal{N}(v) \cup \{v\}\})$
**end for**
**return** $\{h_v^{(L)}\}$

The Aggregate$(\cdot)$ operation of choice was the graph convolutional operator shown in

[15], which can be seen on a node-level as

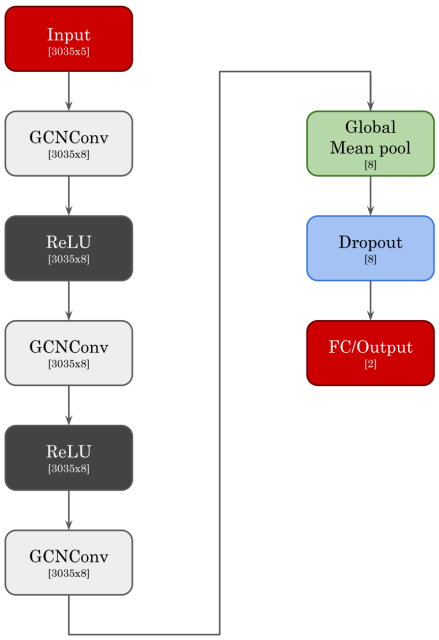
$$h_v^{(l+1)} = \Theta_{(l)}^{\top} \sum_{j \in \mathcal{N}(v) \cup \{v\}} \frac{e_{j,v}}{\sqrt{\hat{d}_j \hat{d}_v}} h_j^{(l)} \tag{4.2}$$

with $\Theta_{(l)}^{\top} \in \mathbb{R}^{n_{l+1} \times n_l}$ being a set of learnable parameters (note how the dimensionality of $h_v^{(l)}$ is not necessarily fixed across layers), $e_{j,v}$ being the weight of the edge between nodes $j$ and $v$ and $\hat{d}_n$ being the weighted "number of neighbours" of node $n$, meaning

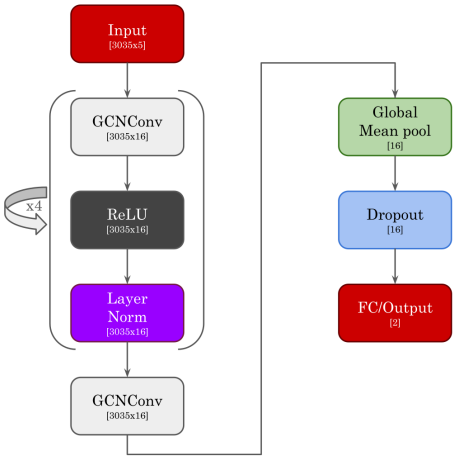
$$\hat{d}_v = 1 + \sum_{j \in \mathcal{N}(v)} e_{j,v} \tag{4.3}$$

where the "1+" is effectively counting for added self-loop connections.

The final architectures can be seen in 4.1, with multiple aggregation operations interleaved with non-linearities, global pooling to reduce the dimensionality, dropout layers and a final fully connected layer to obtain the binary classification.
Different architectures have been applied on different datasets (the difference essentially lying in dimensions - depth and width - and in normalization).
The number under each block represents the dimensionality of the operation result.




**(a)** The architecture of the models trained on the **control**, **smokers** and **non-smokers** datasets.




**(b)** The architecture of the model trained on the **COPD** dataset.

**Figure 4.1.** The final GCN models.

Node features, since naturally absent in a co-expression network, where initialized as random vectors in $\mathbb{R}^5$, with each entry being a standard normal distribution.
The choice of such a compact model, and such a low feature dimensionality, was mainly driven by the low dimensionality of the data: it can be observed that the whole input information consisted of a single real number associated to each edge, i.e. the correlation value.

## 4.2 Model Ensembling

Since the dataset was constructed in a randomized way, it was necessary for the model performance to be independent from the specificity of a single sample. Thus, $N_{models}$ were independently trained and tested; the explanation results must be interpreted as the aggregate explanation of all the trained models.

Refer to the algorithms or formulae of each stage for further details.

## 4.3 GNNExplainer

Explainers are, in general, algorithms that serve the purpose of understanding (potentially a posteriori) what drove the decision of a certain black box AI model. In the context of GNNs, they seek a compact subgraph (of the original input graph) that gets classified in the same way by the model (either on single or multiple classification instances). Such subgraph is then considered to be a good *explanation* for the GNN classification.

In this work, GNNExplainer [16] was chosen. Its optimization process revolves around the following minimization problem

$$\min_{G_s} -\log P_\theta(Y = GNN(G)|G_s) \tag{4.4}$$

where $G$ is the input graph, $G_s$ is the relevant subgraph, $GNN(G)$ is the prediction of the model on the original graph, and $P_\theta(Y = y|G_s)$ is the probability that the output of the GNN, given graph $G_s$, is $y$.
While in equation 4.4 $GNN(G)$ stands for a "hard" prediction (class 0 or 1), in 4.5 the whole output distribution is employed.

The search for $G_s$ is also biased towards small graphs. Practically, GNNExplainer is implemented as a gradient-descent optimization task; it is supervised and its loss function is defined, in this context, as [17]

$$\mathcal{L} = \mathcal{L}_{classification} + \mathcal{L}_{node\ entropy} + \mathcal{L}_{edge\ entropy} + \mathcal{L}_{node\ f.\ size} + \mathcal{L}_{edge\ mask\ size} \tag{4.5}$$

The classification term is just cross-entropy loss, where the target distribution is $GNN(G)$ and the predicted distribution is $GNN(G_s)$. It is insightful to look at the other addends.
The notation will go as follows: $V_s \in \mathbb{R}^{|V_s|}$ is the node mask, where each entry of $V_s$ corresponds to a node in $G_s$. Similarly, $E_s \in \mathbb{R}^{|E_s|}$ is the edge mask, where each entry of $E_s$ corresponds to an edge in $G_s$.
Entropy is meant in an information-theoretic sense, in both cases; it is, essentially, a scaled negative log-likelihood

$$\mathcal{L}_{node\ entropy} = -\frac{\lambda_{V_s\ size}}{|V_s|} \sum_{i=1}^{|V_s|} [(V_s)_i \log((V_s)_i + \epsilon) + (1 - (V_s)_i) \log(1 - (V_s)_i + \epsilon)] \tag{4.6}$$

$$\mathcal{L}_{edge\ entropy} = -\frac{\lambda_{E_s\ size}}{|E_s|} \sum_{i=1}^{|E_s|} [(E_s)_i \log((E_s)_i + \epsilon) + (1 - (E_s)_i) \log(1 - (E_s)_i + \epsilon)] \tag{4.7}$$

with $\lambda_{V_s\ size}$, $\lambda_{E_s\ size}$ being hyperparameters and $\epsilon$ being a stabilization parameter added in order to avoid numerical instability problems.

Node feature size is practically irrelevant in the case of whole-node masking (in other types of task, one may only cover a subset of node features, instead of the entire node); however, it is defined as

$$\mathcal{L}_{node\ f.\ size} = \frac{\lambda_{f.\ size}}{|V_s|} \sum_{i=1}^{|V_s|} (V_s)_i \tag{4.8}$$

where $\lambda_{f.\ size}$ is an hyperparameter.
The interesting part is the last loss addend, defined as

$$\mathcal{L}_{edge\ mask\ size} = \lambda_{edge\ size}|E_s|$$

This is the term pushing for a **compact** explanation. Fine-tuning $\lambda_{edge\ size}$ balances the trade-off between explanation accuracy and size.

It must be remarked that GNNExplainer is instance-based. The optimization process regards a single graph $G$, thus, when multiple runs of explanation are performed, each graph $G_i$ will have its own (a priori independent) explanation subgraph $G_{s_i}$.

In order to assess the explainer's performance, the following metrics were used (note that $GNN(G)$ is meant again as a hard prediction).
Let $N_{samples}$ be the number of explanations performed,
$G_i \in \mathcal{D}_{\mathcal{C}}$,

- Positive fidelity:

$$fid_{+}^{(n)} = 1 - \frac{1}{N_{samples}} \sum_{i=1}^{N_{samples}} \mathbf{1}[GNN(G_i) = GNN(G_i \setminus G_{s_i})] \tag{4.9}$$

- Negative fidelity:

$$fid_{-}^{(n)} = 1 - \frac{1}{N_{samples}} \sum_{i=1}^{N_{samples}} \mathbf{1}[GNN(G_i) = GNN(G_{s_i})] \tag{4.10}$$

where $\mathbf{1}[x = y] = 1 \leftrightarrow x = y$. The superscript $^{(n)}$ refers to the fact that the metric is associated to the $n$-th model.
In fact, the average fidelities will be presented:

$$fid_{+} = \frac{1}{N_{models}} \sum_{n=1}^{N_{models}} fid_{+}^{(n)} \tag{4.11}$$

$$fid_{-} = \frac{1}{N_{models}} \sum_{n=1}^{N_{models}} fid_{-}^{(n)} \tag{4.12}$$

# Chapter 5

# Results and discussion

## 5.1 Model performance

Graph Convolutional Networks, although not the first choice for the task, ended up having excellent performances on the different datasets.
The simpler architecture did not manage to go past the performance of a random shooter for the **COPD** dataset, thus rendering necessary the second architecture; in the end, all the architectures accomplished a test accuracy **above 92%**.
Model performances can be observed in figure 5.1 and table 5.2.

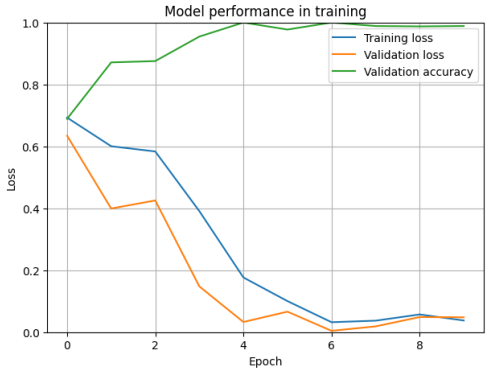
| Hyperparameter/Model task | **COPD** | **Control** | **Smoking** | **Non-smoking** |
|---|---|---|---|---|
| Hidden channels | 16 | 8 | | |
| Batch size | 16 | | | |
| Epochs | 10 | 3 | | |
| Optimizer | Adam, with $lr = 0.1$ | | | |
| Loss f. | Cross entropy | | | |
| Scheduler | StepLR with `step_size=2` and `gamma=0.5` | | | |

**Table 5.1.** Model hyperparameters. "Hidden channels" refers to the dimensionality of vector $h_v^{(l)}$, for $1 \leq l \leq L$.

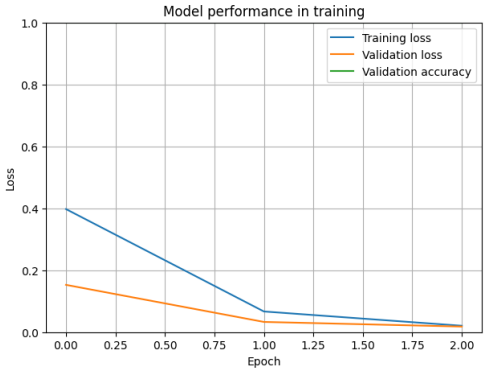
| Metric/Model task | **COPD** | **Control** | **Smoking** | **Non-smoking** |
|---|---|---|---|---|
| Accuracy | 92.42% | 99.77% | 100% | 100% |
| F1 | 92.42% | 99.77% | 100% | 100% |
| Precision | 92.42% | 99.77% | 100% | 100% |
| Recall | 92.42% | 99.77% | 100% | 100% |
| auROC f. | 98.06% | 100% | 100% | 100% |

**Table 5.2.** Model performances (average over each ensemble).

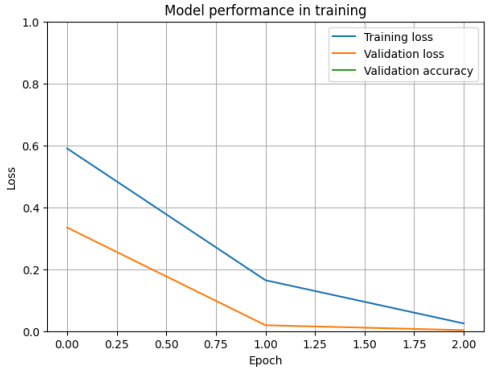
In figure 5.1, validation loss is consistently below training loss. This is probably due to the small dimension of the original dataset.
The almost-perfect results displayed by the model may raise some concern. However, there was no formal sign of overfitting, since these are results on the test set; more importantly, the model predictions were coherent among cohorts (between the 4 dataset types) and their explanation sets seemed to be consistent, to some extent,



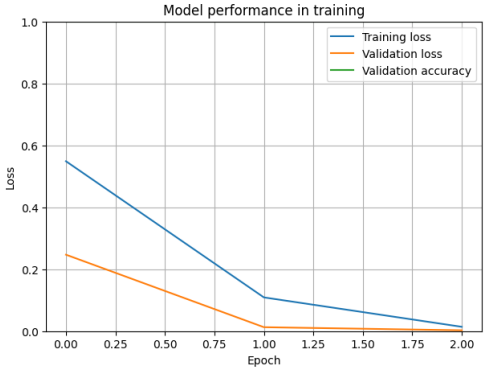

(a) Model trained on the **COPD** dataset.




(b) Model trained on the **control** dataset.




(c) Model trained on the **smoking** dataset.




(d) Model trained on the **non-smoking** dataset.

**Figure 5.1.** Training curves of four example models. Where not visible, accuracy is above the graph.

with the existing biological and medical literature.
In-depth analyses of these two aspects are carried out in sections 5.3, 5.4 and 5.5.

## 5.2 Explainer performance

The explainer's performance can be summarized with the fidelity metrics values in table 5.3 (dataset classes are defined in chapter 5.5).

| Metric/Model trained on | **COPD** | **Control** | **Smoking** | **Non-smoking** |
|---|---|---|---|---|
| $fid_+$ for class 1 | 0.1050 | 0 | 0 | 0 |
| $fid_-$ for class 1 | 0.1725 | 0 | 0 | 0 |
| $fid_+$ for class 0 | 0.1075 | 0.9825 | 0.7775 | 0.6975 |
| $fid_-$ for class 0 | 0.2000 | 0.7950 | 0 | 0 |

**Table 5.3.** Explainer performances (average over each ensemble).

It is possible to interpret these values:

- $fid_- \approx 0$, means that the subgraph $G_{s_i}$ always makes the GNN predict the

same result as $G_i$. Indeed, if the predictions are the same,

$$\mathbb{E}[fid_{-}^{(n)}] = \mathbb{E}[1 - \frac{1}{N_{samples}} \sum_{i=1}^{N_{samples}} \mathbf{1}[GNN(G_i) = GNN(G_{s_i})]$$
$$= 1 - \frac{1}{N_{samples}} \sum_{i=1}^{N_{samples}} 1 = 0$$

and this means that the subgraphs found by the explainer are always sufficient to explain the classification.

- By the same argument, $fid_+ \approx 0$ means that the subgraph $G_s$ is not the only one explaining the GNN classification: $G_i \setminus G_{s_i}$ still contains all the necessary information for the GNN to classify correctly.
- On the other hand, $fid_+ \approx 1$ alludes to the fact that the subgraph $G_s$ contains all the necessary information to predict $GNN(G_i)$, thus the other class must be necessarily predicted, since we are in the context of binary classification.
- $fid_+ \approx 0.5$ is, essentially, the performance of a random shooter, given that the dataset is balanced and the GNN prediction is accurate.
  Indeed, let $x_i \sim Bern(0.5)$; since the dataset is balanced,

$$\mathbb{E}[fid_{+}^{(n)}] = \mathbb{E}[1 - \frac{1}{N_{samples}} \sum_{i=1}^{N_{samples}} \mathbf{1}[GNN(G_i) = x_i)]] = 0.5$$

  where the $x_i$'s are i.i.d; thus, a $fid_+$ close to 0.5 is an hint of having removed both positive and negative genetic markers. These markers may be genes per se or in their connections (the discussion of gene sets as functional units is carried out in section 5.5.2).

These results are encouraging (except for class 0 of the **control** dataset), since carrying out data analysis on the found subgraphs $G_{s_i}$ makes sense given that they always explain the reason behind a specific classification.
However, it must be noted that these subgraphs $G_{s_i}$ may not be unique to each classification (in the case of $fid_+ \approx 0$) .

## 5.3 Analysis of explanation graphs

It is now possible to extract information from the various $G_s$'s found by the explainer, and to compare them across the various classes.
The explainer essentially assigns a "weight" or "relevance" to each node, or gene; formally, it is a multivariate Bernoulli distribution

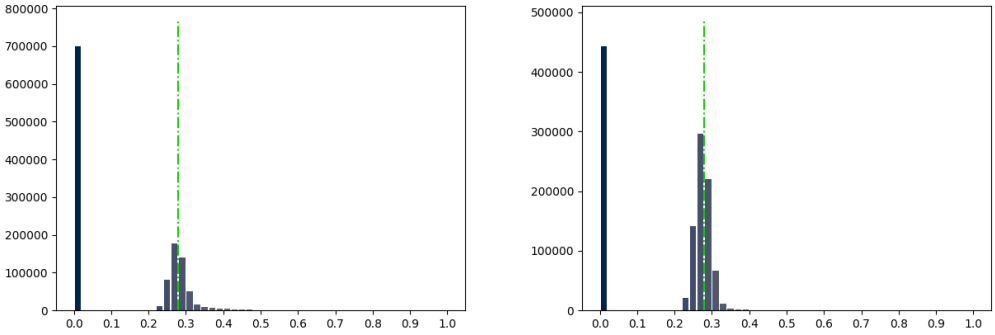
$$P_{\mathcal{G}}(G_s) = \Pi_{v \in G_s} w_v$$

with $\mathbf{1}[v \in G_s^*] \sim Bern(w_v)$, meaning that the presence of $v$ in the "real" explanation set $G_s^*$ is a Bernoullian r.v.; the distribution of such $w_v$'s will be considered. Clearly, $\forall v \notin G_s, w_v = 0$.

The weight distributions for the 8 $\mathcal{D}_{\mathcal{C}}$'s will now be analysed, by reporting the

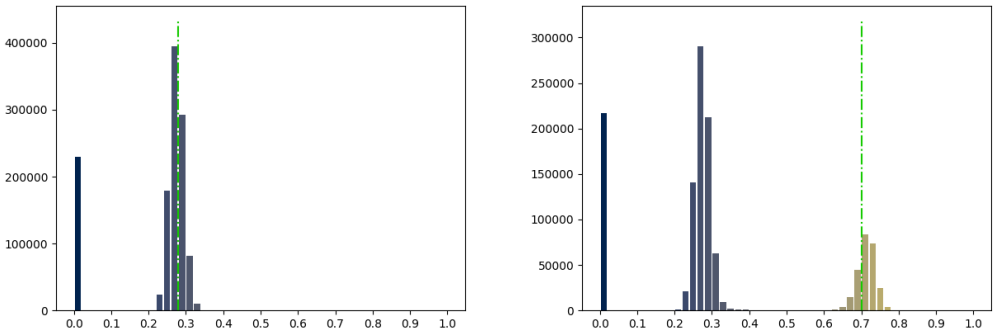
distributions of each pair of classes for the 4 datasets. The next histograms will be computed as follows.
Let $[b_i, b_j)$ be a bin,
$N_{samples} = 20$ be the number of explanations performed,
$G_\ell \in \mathcal{D}_\mathcal{C}$,
$w^n_{v,G_\ell}$ be the weight of node $v$ in the explanation of network $G_\ell$ for the $n$-th model:

$$weight_height(b_i, b_j)_{\mathcal{D}_\mathcal{C}} = \sum_{n=1}^{N_{models}} \sum_{\ell=1}^{N_{samples}} \sum_{v \in G_\ell} \mathbf{1}[b_i \leq w^n_{v,G_\ell} < b_j]$$



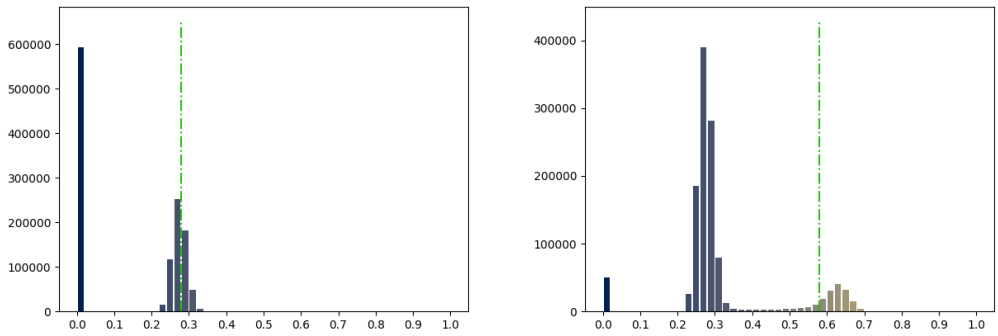

**(a)** Class 1: smoker COPD. **(b)** Class 0: non-smoker COPD.

**Figure 5.2.** Weight distribution for the model trained on the **COPD** dataset.



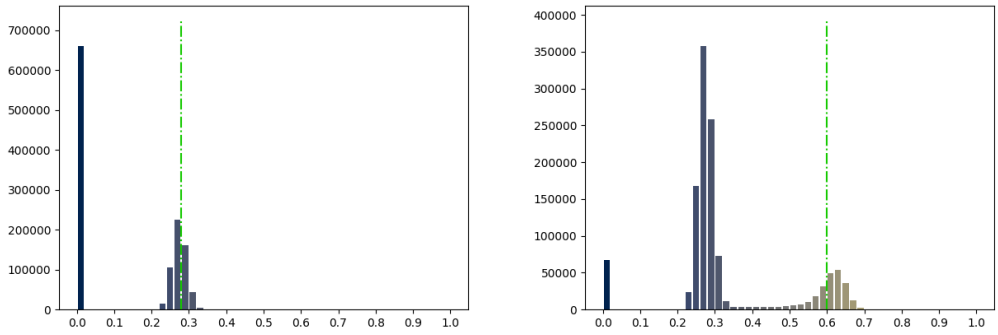

**(a)** Class 1: smoker controls. **(b)** Class 0: non-smoker controls.

**Figure 5.3.** Weight distribution for the model trained on the **control** dataset.




**(a)** Class 1: smoker COPD. **(b)** Class 0: smoker controls.

**Figure 5.4.** Weight distribution for the model trained on the **smokers** dataset.




**(a)** Class 1: non-smoker COPD. **(b)** Class 0: non-smoker controls.

**Figure 5.5.** Weight distribution for the model trained on the **non-smokers** dataset.

In each of the graphs, where present, the green line signals which weight can be considered as a cut-off point, by setting

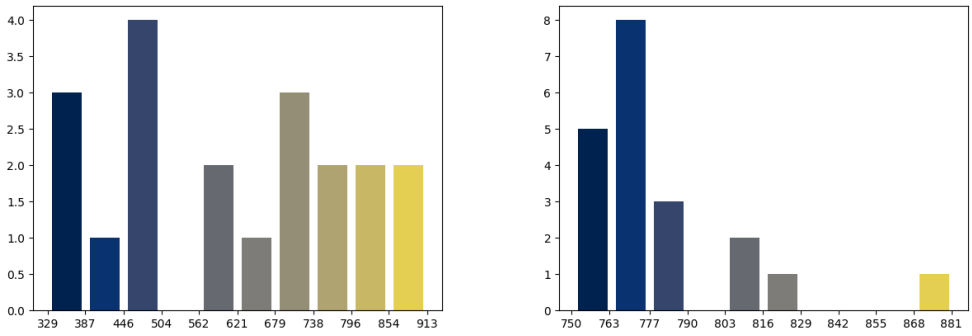
$$weight_cut_{\mathcal{D}_\mathcal{C}} = \arg\min_{w}(CDF[weight_height_{\mathcal{D}_\mathcal{C}}](w) \geq 0.9) \tag{5.1}$$

$CDF[weight_height_{\mathcal{D}_\mathcal{C}}]$ being the cumulative distribution function associated to the (normalized) distribution $weight_height_{\mathcal{D}_\mathcal{C}}$. The index $\mathcal{D}_\mathcal{C}$ is needed since it is relative to a specific cohort in each specific dataset.

The distribution of explanation sizes can be explored; in order to exploit the weight information, such sizes were calculated by considering the genes above the weight cut-off.
Let $[b_i, b_j)$ be a bin,
$N_{samples}$ be the number of explanations performed,
$G_\ell \in \mathcal{D}_\mathcal{C}$,
$w^n_{v,G}$be the weight of node $v$ in the explanation of network $G$ for the $n$-th model:

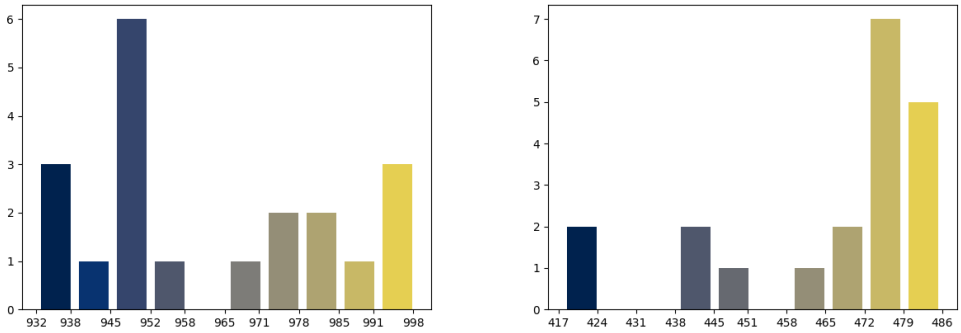
$$size_n(G_\ell)_{\mathcal{D}_\mathcal{C}} = \sum_{v \in G_\ell} \mathbf{1}[w^n_{v,G_\ell} > weight_cut_{\mathcal{D}_\mathcal{C}}] \tag{5.2}$$

$$size_height(b_i, b_j)_{\mathcal{D}_\mathcal{C}} = \sum_{n=1}^{N_{models}} \sum_{\ell=1}^{N_{samples}} \mathbf{1}[b_i \leq size_n(G_\ell)_{\mathcal{D}_\mathcal{C}} < b_j] \tag{5.3}$$



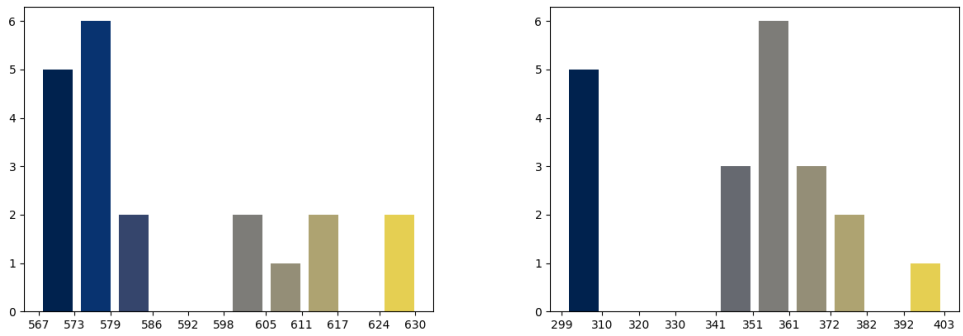

**(a)** Class 1: smoker COPD. **(b)** Class 0: non-smoker COPD.

**Figure 5.6.** Explanation size distribution for the model trained on the **COPD** dataset.



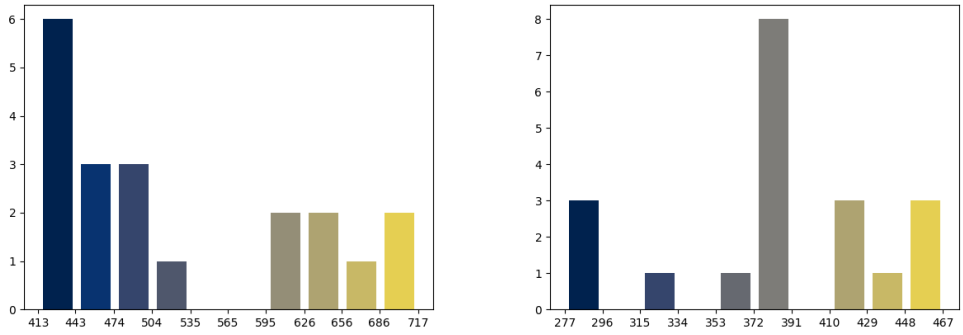

**(a)** Class 1: smoker controls. **(b)** Class 0: non-smoker controls.

**Figure 5.7.** Explanation size distribution for the model trained on the **control** dataset.




**(a)** Class 1: smoker COPD. **(b)** Class 0: smoker controls.

**Figure 5.8.** Explanation size distribution for the model trained on the **smokers** dataset.




**(a)** Class 1: non-smoker COPD. **(b)** Class 0: non-smoker controls.

**Figure 5.9.** Explanation size distribution for the model trained on the **non-smokers** dataset.

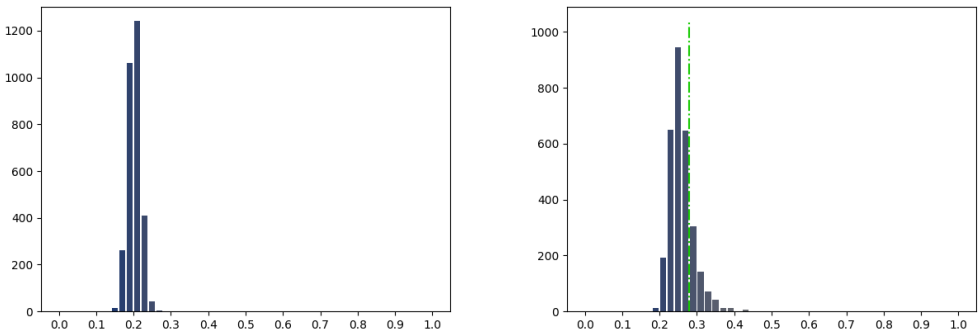
All cohorts have relatively compact explanations.

It is also interesting to look at which relevant (i.e., above cut-off) genes explain many networks. For each class, the distribution of gene frequencies across many explanations will be considered.
Let $[b_i, b_j)$ be a bin,
$N_{samples}$ be the number of explanations performed,
$v \in G_\ell$,
$w^n_{v,G_\ell}$ be the weight of node $v$ in the explanation of network $G_\ell$ for the $n$-th model:

$$count(v, \mathcal{D}_\mathcal{C}) = \frac{1}{N_{models} \cdot N_{samples}} \sum_{n=1}^{N_{models}} \sum_{\ell=1}^{N_{samples}} \mathbf{1}[w^n_{v,G_\ell} > weight_cut_{\mathcal{D}_\mathcal{C}}] \quad (5.4)$$

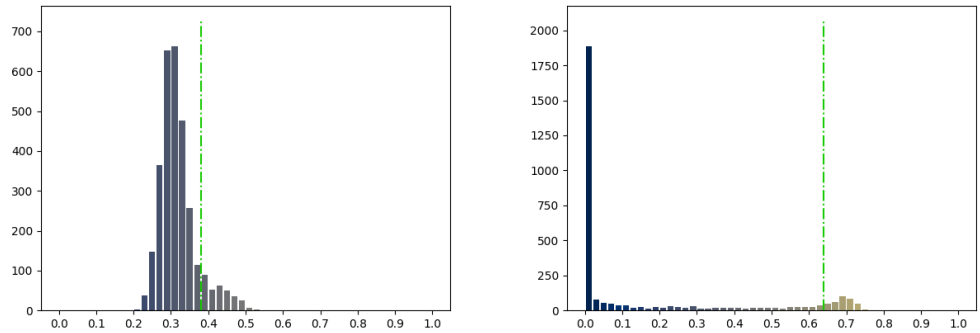
$$count_height(b_i, b_j)_{\mathcal{D}_\mathcal{C}} = \sum_v \mathbf{1}[b_i \leq count(v, \mathcal{D}_\mathcal{C}) < b_j] \quad (5.5)$$

Note how it is possible to write $v$, without specifying the considered $G_\ell$, since every co-expression network contains the same 3035 genes.
This last quantity will be referred to as **gene count distribution**.



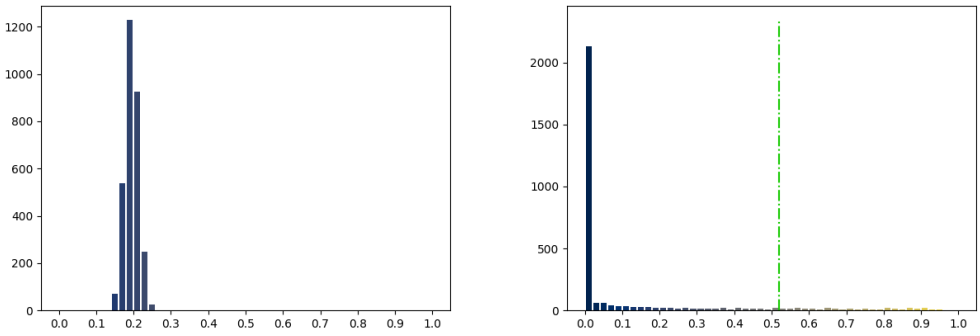

**(a)** Class 1: smoker COPD. **(b)** Class 0: non-smoker COPD.

**Figure 5.10.** Gene count distribution for the model trained on the **COPD** dataset.



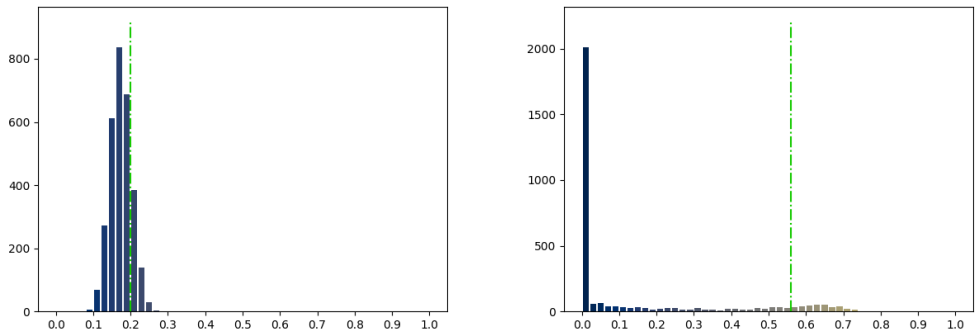

**(a)** Class 1: smoker controls. **(b)** Class 0: non-smoker controls.

**Figure 5.11.** Gene count distribution for the model trained on the **control** dataset.




**(a)** Class 1: smoker COPD. **(b)** Class 0: smoker controls.

**Figure 5.12.** Gene count distribution for the model trained on the **smokers** dataset.




**(a)** Class 1: non-smoker COPD. **(b)** Class 0: non-smoker controls.

**Figure 5.13.** Gene count distribution for the model trained on the **non-smokers** dataset.

The graphs show again the patterns of the weight distribution, and similarly, a green line indicates where the gene count cut-off should be considered:

$$count_cut_{\mathcal{D}_\mathcal{C}} = \arg\min_w (CDF[count_height_{\mathcal{D}_\mathcal{C}}](w) \geq 0.9) \tag{5.6}$$

with $CFD[\cdot]$ defined as before.

The meaning of cutting off is clearer now: it makes sense to cut off distributions following (what resembles) a power law distribution, where genes on the right tail are present in all explanations. On the other hand, there is no dominant element (in this case, gene) in the Gaussian distribution of figures 5.10a and 5.12a, which can be linked with the weaker connection between genetics and COPD for smokers.
It must be noted that this interpretation is not perfect, since, as it can be seen in figures 5.10b and 5.13a, there were explanation sets where the distribution was different, but genetics was still likely to be relevant (given the results of the next sections).

## 5.4 Numerical comparison of relevant gene sets

At this point, the following hypotheses can be formulated:

1. There might a set of genes which guarantees negativity, and this set is (at least partially) shared across smoker and non-smoker controls. It will be called **negativity** set.

2. There is a set of genes which triggers positivity to COPD, without any influence from lifestyle; it will be called **positivity** set, and it is distinct from the negativity set.

3. COPD is correlated with smoking, regardless of the presence of the positivity set, but requiring the absence of the negativity set.

Since different models were trained, different negativity and positivity sets have been found, and need to be compared. The consistency of these sets among the different models will be shown: it is another indicator of their capturing the underlying biological and medical features, rather than spurious correlations.

### 5.4.1 Negativity set

If the negativity set exists, it must be shared between smoker and non-smoker control patients. Thus, a comparison of the gene count distributions of smoker and non-smoker controls is due. Ideally, the frequency of a certain gene across multiple explanations (considering only relative genes) **should be the same for both sets**. The distribution of the difference was then calculated, by subtracting the gene count distributions, cut again on $count_cut_{\mathcal{C}}$; this measure will be called **count difference**.
Let $[b_i, b_j)$ be a bin,
$\mathcal{D}_{\mathcal{C}}$, $\mathcal{D}_{\mathcal{C}'}$ be two sets of negative patients,
$count(v, \mathcal{D}_{\mathcal{C}})$ defined as before,

$$rel(v, \mathcal{D}_{\mathcal{C}}) = \mathbf{1}[count(v, \mathcal{D}_{\mathcal{C}}) \geq count_cut_{\mathcal{D}_{\mathcal{C}}}] \tag{5.7}$$

$$\Delta(v, \mathcal{D}_{\mathcal{C}}, \mathcal{D}_{\mathcal{C}'}) = 2 * \frac{|count(v, \mathcal{D}_{\mathcal{C}}) - count(v, \mathcal{D}_{\mathcal{C}'})|}{count(v, \mathcal{D}_{\mathcal{C}}) + count(v, \mathcal{D}_{\mathcal{C}'})} \tag{5.8}$$

$$count_dif(v, \mathcal{D}_{\mathcal{C}}, \mathcal{D}_{\mathcal{C}'}) = \Delta(v, \mathcal{D}_{\mathcal{C}}, \mathcal{D}_{\mathcal{C}'}) \times rel(v, \mathcal{D}_{\mathcal{C}}) \times rel(v, \mathcal{D}_{\mathcal{C}'}) \tag{5.9}$$

$$count_dif_height(b_i, b_j, \mathcal{D}_{\mathcal{C}}, \mathcal{D}_{\mathcal{C}'}) = \sum_{v} \mathbf{1}[b_i \leq count_dif(v, \mathcal{D}_{\mathcal{C}}, \mathcal{D}_{\mathcal{C}'}) < b_j] \tag{5.10}$$

In other words, it is a histogram of the relevant ($\geq count_cut_{D_C}$) genes' count difference, normalized. Results are shown in figure 5.14.

It is also possible to assess the similarity of the negativity sets (discarding distributional information and only considering genes counted more than $count_cut_{\mathcal{D}_{\mathcal{C}}}$) using Boschloo's test [18], which is based off Fisher's exact test. Fisher's test applies on $2 \times 2$ contingency tables, which in the case of set overlap can be depicted as follows:

| | $X \subseteq A$ | $X \not\subseteq A$ |
|---|---|---|
| $X \subseteq B$ | $\lvert A \cap B \rvert$ | $\lvert \overline{A} \cap B \rvert$ |
| $X \not\subseteq B$ | $\lvert A \cap \overline{B} \rvert$ | $\lvert \overline{A} \cap \overline{B} \rvert$ |

When marginal counts are fixed (i.e. $|A|$ and $|B|$ are given), it tests against the null hypothesis that the two categories are independent. It is possible to see that:

- Out of $N$ possible of genes, $|A \cup B|$ need to be sampled
- Out of $|A \cup B|$ genes, exactly $|\bar{A} \cap B|$ must belong to B but not A
- Out of $|A \cup B|$ genes, exactly $|A \cap B|$ must belong to the intersection
- The remaining cells of the table are exactly determined by the previous statements

Thus, under these hypotheses, the distribution of table values (considered collectively as a random variable) is hypergeometric, and it is possible to calculate how "extreme" the table is: this is the $p$-value of the test.

Fisher's exact test is overly conservative for the considered setting, since the marginals counts are not fixed (there is no assumption on the explanation sizes), which leads to an undercounting of the possible tables and higher $p$-values. Boschloo's test was

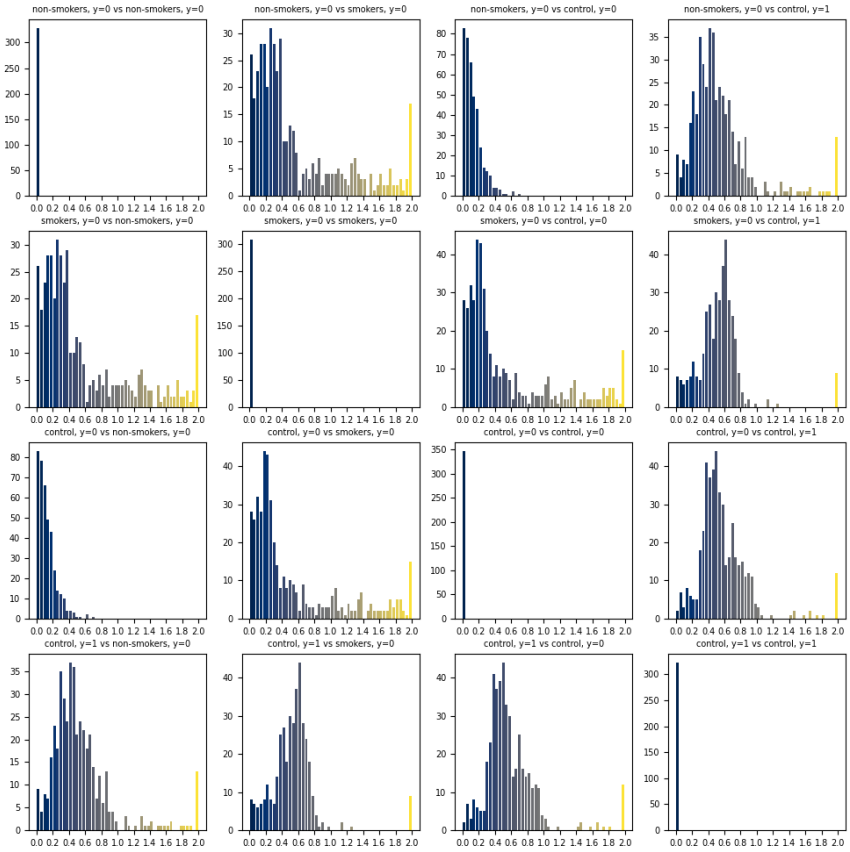



**Figure 5.14.** The count difference among the 4 negativity sets. Note how all distributions are skewed to the right, meaning a similarity between the negativity sets.

considered more appropriate: the fundamental principle is the same as Fisher's exact test, as it is designed to deal with $2 \times 2$ contingency tables, but the only fixed value is the total $N$.
The $p$-value of Boschloo's test, for all the intersections of the four $\mathcal{D}_{\mathcal{C}}$'s, was well below $10^{-10}$, thus making the null hypotheses rejectable at (practically) any confidence level.

These two different approaches showed that the difference between the negativity sets obtained by models trained on different datasets is statistically negligible, thus a negativity set may exist.

### 5.4.2 Positivity set and correlation with smoking

In a similar fashion, it would be logic to see a positivity set. This would be particularly relevant, since, if it can be found in non-smoker COPD networks, but not in smoker COPD networks, it would mean that the latter class is positive to COPD because of factors external to genetics and common to all the cohort's patients, such as smoking itself.
Indeed, the cohort of smokers positive to COPD, both for the **COPD** and **smokers** models, did not have consistently relevant genes, and their correct classification may only be due to the binariness of this classification task.

In figure 5.15, the count difference (as defined in the previous section) of the two cohorts of non-smokers positive to COPD is shown, comparing the explanation sets for the **COPD** model with the one for the **smokers** model.

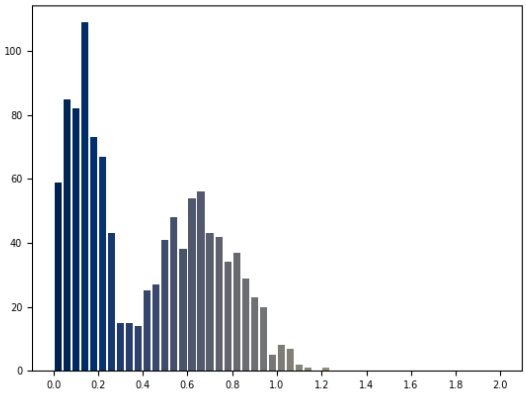



**Figure 5.15.** The count difference among the 2 non-smoker positivity sets.

Even in this case, the $p$-value of Boschloo's test is low ($< 10^{-5}$) and their similarity is statistically significant.

It is also possible to see how different the positivity and the negativity sets are in figure 5.16. The least clear count differences are the two on the last column on the right, as they would seem to show a certain degree of similarity. However, those comparisons involve the explanations of the **control**-trained model, which were shown to be less reliable in section 5.3.

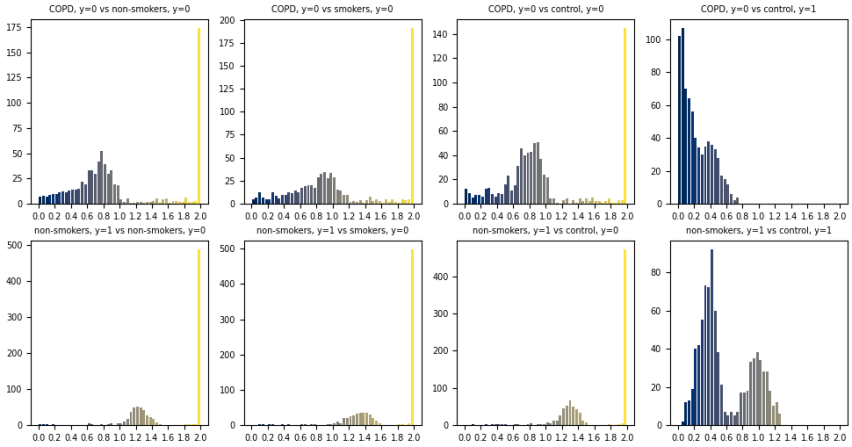



**Figure 5.16.** The count difference between positivity and negativity sets.

## 5.5 Enrichment of relevant gene sets

Once the existence of these positivity and negativity sets was established for the proposed model, it was possible to check whether such gene sets are already known, in the biological literature, for being correlated with COPD. This process is called functional enrichment analysis, and it was performed separately for the positivity and negativity sets; such analysis was carried out using Enrichr [19] [20] [21].

Given a set of genes, Enrichr formulates a number of hypotheses that are tested against various databases; those considered here are DisGeNet [22], Reactome [23] and KEGG [24] [25] [26].
DisGeNet is a gene-disease association database, with data integrated from expert curated repositories, genome-wide association study catalogues, animal models and the scientific literature [22].
Reactome and KEGG are pathway databases. A pathway, in biology, is a series of actions among molecules in a cell that leads to a certain product or a change in the cell. It can trigger the assembly of new molecules, turn genes on and off, or cause the movement of a cell.

Enrichr hypothesis testing is based off Fisher's exact test for $2 \times 2$ contingency tables (explained in section 5.4.1), and the adjusted p-values that are reported are corrected with the Benjamini-Hochberg method.
The Benjamini-Hochberg method is a statistical procedure that binds the false discovery rate (falsely rejected null hypotheses) in multiple hypothesis testing, defined as

$$FDR = \mathbb{E}\left[\frac{\text{False positives}}{\text{Positives}}\right] \tag{5.11}$$

and the procedure can be seen in algorithm 3. It controls which hypotheses get rejected in such a way that $FDR \leq q$. This was rendered necessary by the fact that Enrichr simultaneously tests hypotheses regarding many diseases or pathways at once.

**Algorithm 3** Benjamini-Hochberg Procedure

**Input:** $q$, the control level of the $FDR$; $M$ null hypotheses
  **for** $m = 1 \dots M$ **do**
    $p[m] \leftarrow$ p-value for $H_{0m}$
    $p'[m] \leftarrow$ p-value for $H_{0m}$
  **end for**
  Sort $p'$ in ascending order
  $L \leftarrow \max\{i \mid p'[i] < \frac{q}{M} i\}$
  Reject all $H_{0i}$ s.t. $p[i] \leq p'[L]$

### 5.5.1 Positivity set

By comparing the found positivity sets in the DisGeNet database, an adjusted $p$-value below 0.05 was found for both positivity sets ($< 0.001$ for class 1 of **non-smokers** and 0.032 for class 0 of **COPD**). Table 5.4 shows the intersection between the explainer's positivity set and DisGeNet's COPD-associated genes.

Precaution is always needed when drawing conclusions. This result, by no means, excludes the possibility that the model caught spurious correlations; in no way it was assessed that the chosen architecture accurately models the biological processes behind COPD; yet, it poses statistically significant biological grounds for the explanation sets.

### 5.5.2 Negativity set

The negativity sets considered were the ones from the **smokers** and **non-smokers**-trained models, as the explanation performances for the **control**-trained model were not deemed sufficient (section 5.2). As it can be seen in figure 5.14, these negativity sets show great similarity, and the next considerations will regard hypothesis tests rejected or not rejected by both of these explanation sets.

No significant correlation with COPD was found in DisGeNet. This may be due to a greater attention, in the various studies, posed to the genes causing the disease, rather than the protection mechanisms; on top of this, protection mechanisms may be more generic, and not correlated exactly with a single disease.

However, it was still possible to look after pathways. The following pathways have been found under a confidence level of 95%.
The first pathway was receptor tyrosine kinases, a group of affine cell surface receptors which mediates cell-to-cell communication. Kinase inhibition, and in particular tyrosine kinase inhibition has been shown to reduce airway remodelling due to chronic pulmonary diseases, such as COPD [27].
Another pathway that was found (even though for negative non-smokers only) is the vascular endothelial growth factor (VEGF), which is an essential growth factor for vascular endothelial cells. It preserves the homeostasis of the alveolar components (located in lungs) and it is associated to COPD, as it is upregulated in the non-emphysematous variant of COPD (the one considered in this work) [28] and its inhibition, again, moderates airway remodelling [27].
A last notable pathway was phosphoinositide 3-kinase (PI3K), a signalling pathway involved in cell cycle regulation, whose hyperactivation blocking has been associated to better clinical outcomes in COPD patients [29].

| Class 1 of non-smokers | Class 0 of COPD |
|---|---|
| SERPINA1 | NRP1 |
| CFH | SERPINE2 |
| NCF4 | SRC |
| ITGB2 | TNF |
| PITPNA | MPO |
| PTEN | FBLN5 |
| ADM | CXCL5 |
| PARK7 | ICAM1 |
| IL1RAP | NT5E |
| COX5A | KLRC4-KLRK1 |
| CTSS | CCL2 |
| TBK1 | PDE4A |
| CASP3 | PTGDS |
| TIMP2 | TNFRSF4 |
| HMOX1 | PLXNA4 |
| MAPK1 | PDGFRB |
| CD14 | PTPN1 |
| CD33 | CD74 |
| IL13RA1 | NQ01 |
| CAP1 | GSTM1 |
| CD74 | MIA |
| CREBBP | PRKCB |
| RRM1 | ASCC1 |
| SMAD3 | SPHK1 |
| STAT1 | PDE4D |
| GSS | IL18 |
| ANXA11 | SUMF1 |
| MIF | CHIT1 |
| MMP9 | CXCL10 |
| CS | DLG2 |
| ADAM19 | PPARA |
| TGFBR3 | Ra SIGLEC8 |
| HCK | KLKB1 |
| VCAN | |
| SELL | |
| DDIT3 | |
| ERCC1 | |
| CD68 | |
| S100A9 | |
| TP53 | |

**Table 5.4.** Genes in the intersection of the positivity sets with the COPD gene database from DisGeNet.

# Chapter 6

# Conclusions and possible future work

The methodology proposed was successful both in accomplishing the classification task and getting compact and reliable explanations for all the datasets (with the exception of **control**), in a way which leaves open the possibility that the model actually captured the real motivating genetic influence (when present).
The results of the model were often coherent among themselves. The set of genes associated with positivity showed the strongest correlation with known COPD-associated genes; the set of genes associated with negativity still suggested that the model was not merely distinguishing non-random (positive) from random (negative) graphs. It may actually have given more importance to genes involved in different defence mechanisms preventing the arising of the disease. Randomness seemed, instead, to be more characteristic of the genetics of smokers affected by COPD, a notion that reduces the importance of genetics for such cohort and hints at the well-known correlation of COPD with smoking per se.

It must be noted that all of this was the result of a deep learning pipeline that was not, in any way, informed of the biological task (by the exception, perhaps, of biasing the explainer towards compactness), but that still succeeded in capturing biologically relevant information found with completely different methodologies.
This underlines the possibility of developing gene-discovery methods that are totally unaware of the actual task, of the expected explanation subgraph properties and of the known pathogenetics even for complex diseases.

The possibilities for expansion and improvement are many.
The low explainer performances on the **control** dataset require further investigation in order to understand what actually drives the model classifications. On an architectural side, a multi-omic approach could be attempted, in order to have a higher availability of data and to enable the possibility of more expressive models. For example, proteomic data could be integrated.
The use of correlation networks already poses a strong inductive bias on the model: a transductive approach could remove it, by modelling the problem as a bipartite graph (having genes on one side and patients on another). On the opposite path, the one posing more biases, an approach similar to the one proposed in [9] could be attempted, even though it would need to get adapted for graph classification.
A number of failed attempts on the explainer algorithms were tried during the experiments: for example, biasing the explainer's gene choice by enriching genes with known biological features, or even using different algorithms that try to give

explanations directly for multiple, rather than single, classification instances [30]; two paths that, especially in combination with a richer dataset, could be experimented more carefully to have biologically and/or statistically sounder explanations.
In order to explore further the modelling possibilities of domain-agnostic GNNs in this field, a comparison study could be carried out, by applying the same methodology (including the model-agnostic explainer) with different GNN architectures and exploring which kind of biological features (through the study of pathways, for example) each model naturally captures through its explanation subgraphs.

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